

Can my writing be polished further? When ChatGPT meets human touch

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Recent attention on ChatGPT, a prominent AI language model, highlights its potential in assisting EFL writing. Although ChatGPT's capabilities involve grammar correction, vocabulary enrichment and sentence structuring, its full potential alongside human scaffolding in EFL writing classrooms remains unexplored. This study aims to fill this gap by adopting the cognitive process model of writing as a theoretical framework. Four groups of 19 EFL learners were engaged in think-aloud sessions and interviews. The collaborative writing process with ChatGPT progressed through iterative stages of planning, translating, and reviewing, incorporating various embedded subprocesses. Guidance from the teacher and collaborative work among peers facilitated the effective use of ChatGPT for writing refinement by creating an environment of emotional reassurance and enhancing metacognitive awareness among the learners. The findings emphasise the importance of collaborative work, orchestrated by the teacher and peers in using ChatGPT in the EFL writing class.

Key words: EFL writing, human and AI scaffolding, metacognitive awareness, prompt crafting, writing revision

Introduction

The development of artificial intelligence (AI) has transformed language teaching by providing learners with instant, personalized and interactive experiences (Rusmiyanto *et al.* 2023). ChatGPT, an AI language model, is one of the models that contributes to facilitating instant conversations with language learners and serves as a personal tutor in EFL writing by checking language usage, organization, content, and vocabulary (Godwin-Jones 2021; Dergaa *et al.* 2023; Hong 2023; Schmidt-Fajlik 2023). However, concerns about overreliance on ChatGPT have been raised, such as the risk of incorporating false information, producing unreliable work, and unintentional plagiarism (Dergaa *et al.* 2023). These challenges underscore

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the essential role of teachers who provide necessary scaffolding in guiding learners to use ChatGPT for enhancing their writing competence.

Several studies have emphasized the guidance and support of teachers in using ChatGPT for EFL writing by focusing on the concept of scaffolding: support provided in collaboration with teachers and peers (Vygotsky 1978). Schmidt-Fajlik (2023) conducted a study using ChatGPT as a grammar checker for lower-level Japanese university students. Although the study provided model prompts for students to copy and paste, it did not conclusively show improvements in grammatical accuracy or writing confidence, highlighting the need for more effective scaffolding strategies extending beyond initial interactions with ChatGPT. Similarly, Su, Lin, and Lai (2023) explored the potential use of ChatGPT argumentative essay writing. While the teachers' role in guiding through outline preparation, content revision, and feedback analysis was crucial, the study primarily simulated these processes without directly involving students; thus, it did not provide empirical data on the actual outcomes of student learning. These examples demonstrate the research gap in understanding the full impact of teacher-led scaffolding on student writing skills in using ChatGPT.

Regarding the need for additional scaffolding for EFL learners in using ChatGPT more effectively, Han *et al.* (2023) developed the 'RECIPE' (Revising an Essay with ChatGPT on an Interactive Platform with EFL learners) platform. This platform provided predetermined prompts allowing students to engage in effective conversations with ChatGPT as a writing tutor enabling personalized responses that could enhance students' writing skills. Despite these expectations, even high-proficiency EFL students (B2 to C1 levels based on TOEFL writing scores) experienced difficulty interacting with ChatGPT and required additional personalized teacher guidance. Specifically, these students demanded assistance in developing prompts, understanding feedback from ChatGPT and applying the advice to their writing.

Wang, Li, and Bonk (2024) also discussed the importance of prompt crafting and response interpretation when using ChatGPT in writing. In their study, postsecondary learners (whose L1 was English or others) similarly reported that although developing strategies like designing specific and detailed prompts, pre-prompting with context and knowing how to ask questions to obtain precise responses from ChatGPT would take much time, it is absolutely essential.

Prompt crafting and response interpretation are crucial elements in effectively using ChatGPT. Unlike automated tools, such as Quillbot, DeepL Write, and Trinka that provide immediate corrections and suggestions, manual prompt crafting engages learners in critical thinking and helps them understand the nuances between similar grammatical structures and vocabulary. This process is particularly important for EFL learners who write prompts in English because it encourages deeper cognitive engagement and a better understanding of their uncertainties and learning needs. The ability to manually craft and refine prompts ensures that learners are actively involved in the learning process, leading to more meaningful AI interactions and significant learning outcomes (Wang, Li, and Bonk 2024).

Theoretical framework

Overall, despite the benefits of ChatGPT, the need to guide EFL learners in generating effective prompts and to address the challenges of its use remains unaddressed. Further exploration is required to develop strategies for scaffolding learners' use of ChatGPT in EFL writing classes, fostering metacognitive skills and striking the optimal balance between AI assistance and human guidance. Therefore, this research addresses the following two questions: (1) How do Korean EFL learners incorporate ChatGPT into academic essay writing? and (2) What is the impact of human scaffolding on employing ChatGPT in learners' writing?

The cognitive process of writing model by [Flower and Hayes \(1981\)](#) serves as the foundation for this study including planning, translating (specifically, changing thoughts into words), and reviewing stages. Planning involves generating and organizing ideas under a given topic. Translating indicates transforming these ideas into coherent written language based on the established plan. Reviewing involves evaluating and revising the written text through self-monitoring and incorporating feedback from the teacher and peers. According to [Wei, Shang, and Briody \(2012\)](#), learners moving back and forth through these stages instead of linearly undergoing them reflects the recursive nature of writing as a 'distinctive thinking process' as proposed by Flower and Hayes (1981: 375).

Metacognition—the awareness of one's cognitive processes and ability to effectively navigate them—lies at the core of this process. Metacognition helps learners select appropriate strategies and apply them for specific writing purposes, genres and target readers.

Building on this cognitive framework, the concept of scaffolding is key to enhancing EFL learners' writing skills ([Aydin and Yildiz 2014](#)). Scaffolding involves dynamic support from more knowledgeable others like teachers and peers and is complemented by ChatGPT's assistance. Teacher scaffolding provides strategic guidance and fosters learners' ability to critically evaluate their writing. Peer scaffolding encourages learners to share autonomy and responsibility, facilitate meaningful interactions, model each other's thinking strategies and writing styles, overcome challenges together and co-construct knowledge ([Teng and Huang 2021](#)).

Although human scaffolding fosters a rich environment for cognitive and metacognitive development, ChatGPT scaffolding offers real-time and interactive feedback, which facilitates the practical application of writing skills. It aids learners in idea generation during planning, assists in language refinement during translation and offers feedback for revision during the reviewing stage.

Prompt crafting and response interpretation is integral to this scaffolding approach. Manually crafting prompts encourages learners to deeply engage with their writing tasks by actively participating in the learning process. This method contrasts with the use of automated tools that may lead to passive acceptance of AI-generated responses, ultimately leading to better revisions and improvements in writing skills.

[Figure 1](#) illustrates how EFL learners engaged with human and ChatGPT scaffolding during the three stages of writing. Over a semester, under the instructor's guidance, learners collaboratively refined their writing by

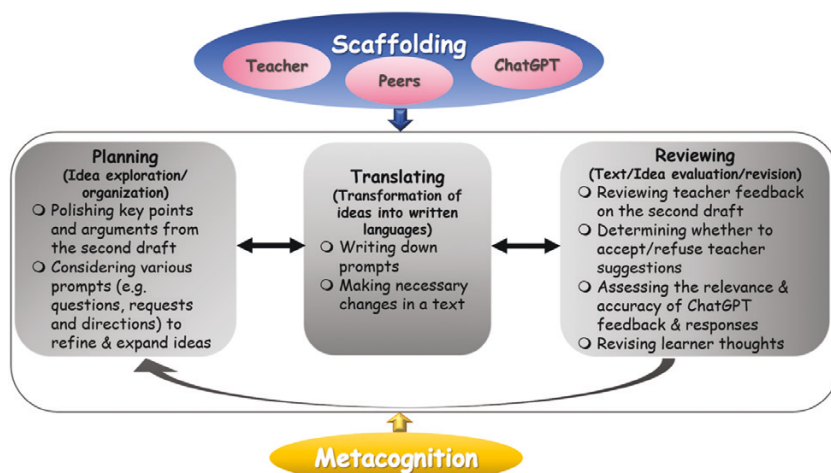


FIGURE 1
Instructional model of
writing with ChatGPT.

The study

Participants

Participants included 19 EFL Korean university students (six males and 13 females), all juniors, majoring in English Language Education. They were enrolled in ‘Logical thinking and writing in English’, a three-credit-hour course, during the 2023 spring semester. They demonstrated intermediate overall English proficiency level (CEFR B1 level), with TOEIC scores of 650–800.

Class procedures

Participants were divided into four groups of four to five students, with careful consideration by the instructor, who also served as the researcher of this study, to ensure complementary dynamics among group members (e.g., grouping students to balance active participation and encourage diverse interaction styles). Throughout the semester, the students wrote three types of essays: narrative, summary and comparison–contrast, following instructions on essay crafting, including paragraph structuring, engaging hooks, clear thesis statements, effective topic sentences, and concise conclusions. The evaluation criteria for each essay type were provided.

ChatGPT was initially recommended for early writing stages like brainstorming and outlining. The instructor proposed that the students use ChatGPT to explore potential similarities or differences between two ideas for brainstorming or for collecting ideas on pointers to include in an introduction. However, most students displayed hesitation and skepticism due to the burdensome nature of writing essays in English and the added stress of formulating and understanding questions in English along with the fear of becoming overly reliant on ChatGPT.

To address these concerns and better prepare the students for effective use of ChatGPT in the writing process, the instructor conducted a focused 30-minute prompt development and response interpretation session. She used example essays and prompts in simulations to demonstrate how

well-crafted questions generate clear answers. This modelling emphasized the importance of engaging introductions, coherent thesis statements and well-constructed topic sentences and equipped the students with skills required to effectively formulate inquiries and eventually interpret ChatGPT's responses to polish their writing.

Following these initial discussions and the prompt development and response interpretation session, the study selected ChatGPT to refine drafts at an advanced stage. Specifically, the consensus was to employ ChatGPT to enhance the final third draft after completing the second draft and receiving feedback from peers and instructor (on the first and second draft, respectively). Therefore, the 15-week course allocated three weeks to use ChatGPT only for the final refinement stage of writing the three types of essays during regular class hours.

Data collection and analysis

The think-aloud (TA) method, recognized for its effectiveness in investigating the cognitive composing process of learners (Hayes and Flower 1981), was used during the final comparison–contrast essay writing. This method requires learners to verbalize and logically organize their thoughts and actions while performing a task, encouraging metacognitive awareness through reflection on writing strategies (Gerova 2022).

Each group underwent a 25-minute TA session which mirrored the structure of regular writing classes performed for the first two essays: narrative and summary. Throughout the semester, the instructor consistently facilitated TA sessions within the class time, focusing on guiding the students to effectively use ChatGPT for draft revisions. Emphasis was placed on developing skills in formulating queries, interpreting feedback and integrating these insights into the revisions. These sessions were conducted in Korean, the students' mother tongue, to reduce their cognitive burden. However, all ChatGPT prompts were required to be in English because engaging with the AI tool in English was considered an integral part of the writing revision process.

Following the TA sessions, a 15-minute interview in the students' L1 was conducted with each group. It covered the past experiences of the students using AI tools in writing, the benefits and challenges of employing ChatGPT and suggestions for more effective integration of ChatGPT in writing classes. After the interviews, for approximately 60 minutes of class time, the students continued to interact with ChatGPT in English to refine their second drafts. However, discussions among them during this process were conducted in L1 to facilitate clearer communication and understanding. This approach was consistent with their method for the previous essays. As a group, they documented their interactions with ChatGPT in a table format with three columns by following the instructor's model portfolio guidelines, including (1) major and meaningful prompts they input, (2) the responses received, and (3) personal reflections, such as points of confusion or insights gained (1 and 2 were recorded in English, whereas 3 was optional and could be in either English or Korean). The data collection procedures are summarized in Table 1.

The TA sessions and follow-up interviews with the four groups were audio-recorded, transcribed and translated into English for analysis. Parts

Data collection-methods	Procedures	Duration	Lg. use
Think-aloud session	- Students (Ss) go through the second draft as a group. - Ss move through the three stages to discuss problematic areas and ways to revise them. - Ss interact with ChatGPT and obtain T support when necessary. - T observe Ss, they articulate their thoughts out loud.	25 min	L1 (discussions) and English (interactions with ChatGPT)
Interview	- The perceptions of Ss of experiences with Chat-GPT use are gathered.	15 min	L1
Further interactions with ChatGPT	- The continuous interactions of Ss with ChatGPT while undergoing the three stages are documented as a portfolio.	60 min	L1 (discussions) and English (portfolio, ≤ 30% L1)

TABLE 1
Data collection
procedures

Findings

Employment of ChatGPT in the writing process

requiring clarifications were member-checked by the students to enhance validity and credibility. After a thorough examination of the transcribed data and the group portfolio data of the students, two major themes emerged: the use of ChatGPT and its outcomes. Then the data were coded by associating each utterance with the corresponding themes regarding the impacts of ChatGPT, revealing two subthemes: the attainment of emotional reassurance and enhancement of metacognitive awareness in writing.

The first research question addressed how the students incorporated ChatGPT into refining writing. Results demonstrated that they primarily used ChatGPT to assess grammar, sentence structure, logical coherence, and subtle nuances of words within their specific writing context. Figure 2 shows how a group of students interacted with ChatGPT to enhance their sentence quality (they wrote about similarities and differences between school teachers and private academy teachers).

The students started from the reviewing stage (Review 1), driven by uncertainties concerning the differences among the words ‘entail’, ‘include’, and ‘have’. By going through the planning, translating, and reviewing stages, they refined their prompts, adding details until they obtained the desired responses. They often noted prompts immediately without planning or they engaged in constant and thorough translation and review (Review 2–3 and Translate 2–3). This process mirrors Flower and Hayes (1981)’s nonlinear cognitive writing model. Numerous subprocesses that highlight the dynamic nature of writing and learning were embedded within this process.

Significantly, the integration of human scaffolding—through the instructor and peer collaboration—played a crucial role. Instead of providing direct responses, the instructor strategically intervened, prompting students to explore in-depth linguistic considerations while peer discussions fostered a collaborative learning environment. This human scaffolding was effectively complemented by ChatGPT, which enabled the students to navigate through their uncertainties more effectively, not only by crafting prompts

FIGURE 2
Cognitive writing
process using
ChatGPT in TA
session 1

(Review 1)
Student 1 Can you understand why the professor recommended alternative words instead of 'have' (referring to the sentence, 'Both jobs have the responsibility of caring for their students')?
Student 2 Oh, I wasn't sure why 'have' is inappropriate.
(Plan 1)
 ...
(Translate 1)
Student 1 (drafting the following prompt: My professor gives me feedback to use 'entail' or 'include' rather than 'have'.) Is this enough, professor?
 ...
Instructor What exactly do you want to know here?
Student 2 Maybe ... specific differences among the three words?
Instructor There you go. Why don't you add one more sentence then?
Student 1 Okay. (adding a sentence: Please tell me the differences when using 'entail' or 'include' instead of 'have'.)
(Review 2 + Translate 2)
Student 1 I'm still not clear about the differences. Professor, can you check what's wrong once again?
Instructor What about reporting what word your group first wrote, and then what words I had suggested. Then you can ask about differences among the three synonyms.
Student 1 Oh, we got it! (writing down the following prompt: I first used the word, 'have' but my professor provided feedback on this sentence using 'entail' or 'include'. I think all three words are synonyms, so tell me about the differences.)
(Review 3)
Student 3 ChatGPT says that 'entail' emphasizes an inherent responsibility, whereas 'include' implies a responsibility within the broader scope of the roles.
Student 2 Considering the emphasis on responsibility, 'include' seems more fitting. What do you think?
(Translate 3)
Student 3 I agree (others nodded in agreement). Let's replace 'have' with 'include' then.

but also by interpreting ChatGPT's responses critically and incorporating them into their writing.

The students highly valued the detailed ChatGPT feedback on awkward, ungrammatical, and illogical parts within their writing, along with clear reasons.

ChatGPT provided detailed clarifications about why we made certain mistakes, which served as a driving force for us to keep asking, refining our writing, and making improvements. (Student 1, Group 1 Interview)

Additionally, the journey to reach their destination included various embedded subprocesses often resulting in moments of fatigue for the students.

We often declared each attempt to be our final version. We said, 'No more changes!' But upon receiving responses from ChatGPT, we realized that there were still some lingering uncertainties in our writing. Sometimes, we felt fatigue and frustration. But the process of gradually uncovering the answers we sought was like a treasure hunt, and it was often enjoyable. (Student 8, Group 2 Interview)

It appeared certain that this involvement of ChatGPT, combined with human scaffolding, fostered a more profound back-and-forth exploration of the planning, translating, and reviewing phases among the students.

The second research question identified the specific outcomes of integrating ChatGPT into human scaffolding in the students' writing process. The focus shifted from the procedural aspects of using ChatGPT to educational and affective outcomes of this integration.

Impact of
ChatGPT
with human
scaffolding in the
writing process

Attainment
of emotional
reassurance in
writing

Although the students collectively engaged with ChatGPT to satisfy their curiosity, they frequently encountered roadblocks that required the instructor's intervention. Figure 3 illustrates the role of the instructor in guiding the students to effectively use ChatGPT, specifically in writing about similarities and differences between texting and calling.

When the students could not continue posing questions to ChatGPT, the instructor assisted them in rephrasing their inquiries. This input enabled them to elicit more satisfactory responses from ChatGPT. Their responses in the subsequent interview are as follows (meaningful parts are underlined):

Student 11 ... to get meaningful answers from ChatGPT, we had to clearly grasp what we were unsure about and what we were seeking. Otherwise, as seen today, our questions would remain unresolved.

Student 10 Absolutely, what was truly impressive was that you (i.e. the instructor) understood precisely what we were uncertain about and stepped in when it was crucial, offering timely help.

Student 14 Exactly!... Having both ChatGPT and you as resources brought a reassuring sense of support. (Group 3 Interview)

As Student 11 mentioned, without a clear grasp of their uncertainties, they could not fully exploit the benefits of ChatGPT. Initially, they found formulating a proper prompt challenging. However, the instructor anticipated the students' needs, understood their challenges and offered timely, relevant assistance, as noted by Students 10 and 14. Wang, Li, and Bonk (2024) also indicated that without effective prompting, learners risk receiving generic responses from ChatGPT, leading to frustration and potential reluctance in continuing using the tool. This implies the importance of developing good questioning skills to elicit meaningful responses. Herein, the instructor's scaffolding helped the students refine

(Review 1)
Student 12 I'm a bit puzzled about why using just 'are' isn't enough in this sentence (i.e. 'The second common characteristic between texting and calling is they are non-face-to-face communication methods')?
(Plan 1)
...
(Translate 1)
Student 11 (jotting down the following prompt: I wrote, 'The second common ...' But, my professor said that the verb 'are' is not appropriate. How can I revise it?)
(Review 1)
Student 10 It offered a revision (i.e. 'Another shared characteristic of texting and calling is they both involve non-face-to-face communication methods.') by replacing 'are' with 'involve,' but it didn't explain why.
(Translate 2)
Student 11 Let's ask once more (writing down the following prompt: But my professor said 'are' is not appropriate. Why is our first sentence wrong?).
(Review 2)
Student 14 It says that the original sentence might be redundant or awkward, but I don't quite understand.
(Plan 2)
Student 12 Maybe we should ask about it the professor. Professor, could you lend us a hand?
Instructor Certainly. Following ChatGPT's suggestion to include the word 'involve' helps clarify that the common characteristic being discussed pertains to their function as non-face-to-face communication methods. You can ask ChatGPT once more why 'are' falls short in providing the necessary detail about the common traits of texting and calling.

FIGURE 3
Cognitive writing process
using ChatGPT in TA
session 3

their prompts and interpret ChatGPT's feedback more effectively, providing them with emotional reassurance in their writing process.

Meanwhile, not only the guidance of the instructor but also that of their peers created a supportive and encouraging environment. [Figure 4](#) illustrates a part of a portfolio submitted by Group 4. They attempted to examine the suitability of their essay title. The prompts were indicated in their words with noteworthy parts marked in bold and underlined.

During the collaborative writing process, a consistent tendency of rephrasing and refining prompts for ChatGPT was observed. Rather than relying solely on ChatGPT's guidance, Group 4 employed a distinctive approach by offering specific guidance to ChatGPT (#5 in [Figure 4](#)). Through extensive discussions, they recognized the value of having their peers for enhanced reliability.

Student 16 ChatGPT interactions sometimes brought challenges, making me want to quit asking questions. But our group's encouragement pushed me to persevere, resulting in better responses from ChatGPT.

Student 18 I totally agree. I usually lacked standout ideas in writing, but collaborating with others in using ChatGPT sparked fresh and captivating thoughts.

[Wang, Li, and Bonk \(2024\)](#) emphasized that peer collaboration opportunities in writing could strengthen self-directed learning through AI by reducing excessive dependency on it and enhancing cognitive and emotional engagement. Similarly, this study found that the students' collaborative work with ChatGPT stimulated their autonomy and active

FIGURE 4
Sample of
collaborative work
with ChatGPT

Prompts	ChatGPT Responses	Reflections
1. Please recommend a theme for this essay.	1. ... a suitable theme for this essay could be: 'Exploring the Similarities and Differences between Movies and Dramas as Engaging Hobbies' ...	1. Our intention was to examine how ChatGPT derives a theme for our essay. It explained the theme by thoroughly analysing our essay. <u>So, we felt like we did a pretty good job.</u>
2. Can you make it shorter?	2. Comparing Movies and Dramas: Similarities and Differences as Hobbies	2. However, the answer was too long, so we asked ChatGPT to make it shorter.
3. It seems to be too typical. Can you recommend another one?	3. The Allure of Movies and Dramas: Exploring Engaging Home Entertainment	3. The proposed theme seemed quite typical. We were hoping for something more unique.
4. 'Distinguished features of dramas and movies': Is this title appropriate for this essay?	4. ... it might be beneficial to consider adding a bit more context ... for example, 'Exploring the Distinguished Features of Dramas and Movies'.	4. We dropped the word, 'theme' and asked ChatGPT about the appropriateness of our essay title.
5. Consider the following points when giving your response: 1) ... 2) <u>Say 'yes' or 'no' if the title, "Distinguished feature of dramas and movies" is appropriate or not ...</u>	5. Yes, the title 'Distinguished Feature of Dramas and Movies' is appropriate for this essay.	5. Getting a satisfactory response from ChatGPT was a bit challenging. But, <u>when it approved our essay title, we felt a sense of relief and recognition.</u>

Enhancement of metacognitive awareness in writing

involvement. They exchanged ideas and reflections, modelled their peers' thinking strategies and writing styles, negotiated meanings and collectively constructed language knowledge to overcome challenges. This approach, as underscored by Flower and Hayes (1981), elevated the performance of the students by enriching their cognitive writing process.

The students progressively adapted their approach to querying ChatGPT, gradually reducing the time they spent replying on guidance from their instructor. This shift was not merely a pause but rather marked the initiation of their quest for more profound learning experiences. Following the instructor's lead, the students in Group 3 (Figure 3) adeptly rephrased the prompts, effectively eliciting the desired responses from ChatGPT. However, their eagerness to explore led them to unexpectedly pose an unplanned question. The continuation of their writing process is illustrated in Figure 5.

The students willingly dug deep, critically analysing the responses provided by ChatGPT and posing a follow-up question. In doing so, the students demonstrated an awareness of their cognitive processes and the ability to adeptly regulate their strategies, indicating an evident elevation in their metacognitive awareness.

The impact of ChatGPT on promoting the students' metacognitive awareness can also be observed from the portfolio created by Group 2 (#2 in the reflection column of Figure 6).

After Group 2 had finalized their essay writing through interactions with ChatGPT, seeking answers to their primary inquiries, they added questions to assess both content reliability and idea coherence. This reflects their curiosity and desire for a sense of accomplishment. As they cross-referenced ChatGPT's suggested ideas with their writing, they found a considerable agreement between the two (#1 in the reflection column of Figure 6).

As discussed by Teng and Huang (2021), possessing metacognition means understanding when and how to apply specific strategies for learning and problem-solving. They emphasized the importance of scaffolding in fostering the metacognitive abilities of learners. Furthermore, they noted that within the context of using these abilities, collaborative writing not only allows learners to collectively address weaknesses but also facilitates mutual support, ultimately improving academic writing skills.

FIGURE 5
In-depth writing process
using ChatGPT in TA
session 3

(Review 1)

Student 10 It looks like we've got a bit of an unexpected issue here.

Student 11 What's the problem?

Student 12 When I looked at the sentence suggested by ChatGPT, I noticed a mismatch in the format of the topic sentences between body 1 and body 2.

(Plan 1 + Translate 1)

Student 10 How about asking ChatGPT to ensure consistent formatting? (written after group discussion: Considering the first topic sentence, can you help me keep the consistent format and suggest a better second topic sentence?)

(ChatGPT generates the following sentence: 'The second common characteristic shared by texting and calling is that both serve as non-face-to-face communication methods.')

FIGURE 6
Sample of
metacognitive
writing behavior
using ChatGPT

Prompts	ChatGPT Responses	Reflections
1. We're working on an essay that compares backpacking and package tours. Can you share your thoughts on what similarities and differences you find between the two?	1. Certainly! Here are some key commonalities and differences between backpacking and package tours: ...	1. <u>We wanted to check if our essay was written well enough.</u> So, we gave ChatGPT our topic and asked it to point out the similarities and differences. Although it offered many suggestions, <u>starting questions beforehand might have led us to rely heavily on its ideas, making it hard to write in our own style.</u>
2. How can we use the similarities and differences you mentioned to write a point-by-point comparison?	2. Here's an example of a point-by-point comparison between backpacking and package tours using the commonalities and differences mentioned earlier! ...	2. Surprisingly, <u>some of the points were like what we wrote in our essay's body paragraphs. In the end, we felt like our thinking was on the right track because it matched up with what came out.</u>

In essence, introducing ChatGPT into collaborative writing with the support of the instructor and peers helped the students refine prompts and expand their critical evaluation of feedback, which was instrumental in enhancing their metacognitive insights and the quality of their writing revisions. Throughout this process, they experienced a sense of emotional stability and reassurance, recognizing their ability to openly discuss challenges in the study and seek guidance. With the ability to monitor and control their writing processes, the students played an active role and used ChatGPT to confirm ideas and explore deeper curiosity. This led to a sense of relief and satisfaction and eventually enhanced their metacognitive awareness, contributing to overall writing refinement.

Conclusion

The integration of ChatGPT into EFL writing has recently caught educators' attention. Although ChatGPT is renowned for its advantages in enhancing language usage, organizing ideas and content and enriching vocabulary, this study explored its potential beyond these aspects. Specifically, it investigated how Korean EFL students use ChatGPT to develop their writing, following the cognitive process of a writing model. Furthermore, this study examined the impact of collaborative use of ChatGPT with guidance from the instructor and peers, aiming to enhance autonomy and proficiency in using ChatGPT as a writing assistant.

The findings revealed that the students interacted with ChatGPT to plan, translate, and review their work, often in a nonlinear order, engaging in embedded subprocesses. Significantly, the integration of human scaffolding—not just to help with prompt formulation but also to interpret ChatGPT's feedback—played a crucial role. This collaborative approach instilled a sense of patience, involvement, and reassurance among the students. With continued use of ChatGPT, they developed more queries and concerns, improving their metacognitive awareness and writing skills.

The findings suggest a few implications of the use of ChatGPT in EFL writing classrooms. First, human scaffolding is crucial across all proficiency levels. Providing a list of simple prompts for ChatGPT can serve as a

starting point; however, the real value lies in encouraging learners to critically think and refine their queries. Regular, brief, and focused practice sessions thus should be incorporated into class time to help learners articulate their thoughts clearly and adapt prompts based on feedback, fostering an in-depth linguistic understanding.

Additionally, teachers should act more as facilitators of knowledge rather than direct providers of information. By offering strategic and timely guidance, learners can make independent decisions about improving their writing. This method not only helps learners take ownership of their learning process but also encourages critical evaluation and independent cognition, ideal for developing language and metacognitive skills.

Despite these insights, this study has certain limitations. First, the small sample size limits the generalizability of the findings. Future research should consider larger-scale studies with a mixed-methods approach to capture a more comprehensive picture of AI tool usage among EFL learners and include objective measures of writing improvement. Second, although this study focused on ChatGPT, comparing the effectiveness of other AI tools like Quillbot, DeepL Write, and Trinka could provide further insights into their role in enhancing academic writing.

In conclusion, integrating ChatGPT into EFL writing classrooms offers advantages in enhancing learners' writing skills and fostering their metacognitive growth with guidance and support from strategic teachers and collaboration among peers. However, it is important to critically evaluate ChatGPT's responses and be mindful of potential challenges like preventing overreliance on it, which may limit learners' independent creativity. Thus, although ChatGPT offers valuable opportunities for EFL writing, its use should be carefully balanced.

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